

# Prompt Engineering Notes

## Learning Objectives

- Understand Large Language Models (LLMs)
- Learn Prompt Engineering
- Understand Tokens
- Learn Temperature
- Study Prompt Types
- Understand API Keys

## 1. Introduction

Prompt Engineering is the process of writing effective instructions (prompts) for AI models to generate accurate and useful responses. A good prompt helps the AI understand the task, context, and expected output.

Large Language Models (LLMs) like ChatGPT, Gemini, Claude, and Llama rely on prompts to perform tasks such as answering questions, writing code, summarizing documents, translating languages, and generating content.

## 2. What is a Large Language Model (LLM)?

A Large Language Model (LLM) is an AI model trained on a large amount of text data to understand and generate human language.

### Examples

- ChatGPT
- Google Gemini
- Claude
- Llama
- Mistral

### Applications

- Chatbots
- Content Writing
- Code Generation
- Translation
- Email Writing
- Research Assistance
- Customer Support

### Advantages

- Saves time
- Increases productivity
- Supports multiple languages
- Generates human-like text

### Limitations

- Can generate incorrect information
- Depends on prompt quality
- Requires fact-checking for important tasks

### 3. What is Prompt Engineering?

Prompt Engineering is the practice of designing clear and effective prompts to get the best possible output from an AI model.

#### Poor Prompt

*Explain Python.*

#### Better Prompt

*Explain Python programming for beginners with examples, advantages, and real-world applications in simple language.*

The second prompt is better because it includes:

- Target audience
- Topic
- Required sections
- Language style

### 4. Components of a Good Prompt

A good prompt should include:

#### Role

Assign a role to the AI.

*You are a Python instructor.*

#### Task

Clearly define what needs to be done.

*Explain Machine Learning.*

#### Context

Provide background information.

*The audience is MCA first-year students.*

#### Constraints

Specify limitations.

- Maximum 300 words
- Use simple language
- Include one example

## Output Format

Tell the AI how to present the answer.

- Bullet points
- Table
- JSON
- Numbered list

## 5. Tokens

A token is the smallest unit of text processed by an AI model. Tokens may represent words, parts of words, punctuation marks, or symbols.

### Example — Sentence

*I love Artificial Intelligence.*

### Possible Tokens

- I
- love
- Artificial
- Intelligence
- .

### Why Tokens Matter

- API pricing is often based on tokens.
- Every model has a maximum token limit.
- Both the input prompt and the generated response use tokens.

## 6. Temperature

Temperature controls the creativity of the AI's responses.

| Temperature | Output                               |
|-------------|--------------------------------------|
| 0.0         | Very factual and consistent          |
| 0.3         | Mostly factual                       |
| 0.7         | Balanced                             |
| 1.0         | Creative                             |
| Above 1.0   | Highly creative but less predictable |

Example Prompt

Write a slogan for a coffee shop.

| Temperature | Output                             |
|-------------|------------------------------------|
| 0.2         | Fresh Coffee Every Morning.        |
| 0.9         | Wake Up to Every Sip of Happiness! |

## 7. Prompt Types

### Zero-shot Prompting

No examples are provided.

*Explain Cloud Computing.*

### One-shot Prompting

One example is provided before the task.

*Input: Apple → Fruit*

*Input: Car → ? Output: Vehicle*

### Few-shot Prompting

Multiple examples are provided. This improves consistency and accuracy.

### Role Prompting

The AI is assigned a specific role.

*You are an HR Manager. Review my resume.*

### Context Prompting

Background information is provided.

*I am preparing for my MCA viva. Explain Decision Trees in simple language.*

### Chain-of-Thought Prompting

The AI is asked to solve the problem step by step.

*Solve the following math problem step by step.*

## 8. Example of an Effective Prompt

*You are an experienced Machine Learning professor. Explain Machine Learning for MCA first-year students. Include: Definition, Types, Advantages, Disadvantages, One real-world example. Present the answer in bullet points and keep it within 400 words.*

This prompt produces a clearer and more structured response than a simple prompt.

## Part 1 Summary

In this section, you learned:

- Introduction to Prompt Engineering
- Large Language Models (LLMs)
- Components of a Good Prompt
- Tokens
- Temperature
- Major Prompt Types
- How to write an effective prompt

## 9. API Keys

### What is an API Key?

An API (Application Programming Interface) Key is a unique code used to authenticate a user or application when accessing an AI service.

It acts like a password and allows applications to communicate securely with AI models.

### Why API Keys are Used

- Authenticate users
- Secure API access
- Track API usage
- Prevent unauthorized access
- Manage billing and quotas

### Example Workflow

*Application → API Key → AI Service → Response*

### Best Practices

- Keep API keys private.
- Never upload API keys to GitHub.
- Store keys using environment variables.
- Rotate keys if they are exposed.

## 10. Prompt Design Best Practices

To obtain accurate responses from AI, follow these guidelines:

### 1. Be Clear and Specific

-  Explain AI.
-  Explain Artificial Intelligence for beginners with examples.

### 2. Assign a Role

*You are a Data Scientist. Explain Decision Trees.*

### 3. Provide Context

*I am preparing for my MCA examination.*

### 4. Specify the Output Format

- Bullet points

- Table
- JSON
- Numbered list

## 5. Set Constraints

- Maximum 300 words
- Use simple language
- Include three examples

## 6. Improve the Prompt

If the output is not satisfactory, refine the prompt by adding more details or constraints.

# 11. Common Prompt Engineering Mistakes

### Mistake 1: Vague Prompt

-  Tell me about Python.
-  Explain Python programming for beginners with examples.

### Mistake 2: No Context

Without context, the AI may generate a generic response.

### Mistake 3: No Output Format

Instead of:

*Compare Python and Java.*

Use:

*Compare Python and Java in a table.*

### Mistake 4: Too Many Tasks

Avoid combining unrelated requests into one prompt.

-  Explain AI, write Python code, summarize a report, and create an email.

Break large tasks into separate prompts.



## 12. Prompt Improvement Exercise

### Version 1

#### Prompt

*Explain Machine Learning.*

#### Output

A short and general explanation.

### Version 2

#### Prompt

*Explain Machine Learning in simple language.*

#### Output

A clearer explanation suitable for beginners.

### Version 3

#### Prompt

*Explain Machine Learning for MCA first-year students. Include its definition, types, advantages, disadvantages, and one example.*

#### Output

A structured answer with relevant sections.

### Version 4

#### Prompt

*You are a Machine Learning professor. Explain Machine Learning for MCA students using simple language. Include: Definition, Types, Advantages, Disadvantages, Real-world example, Comparison table. Limit the response to 500 words.*

#### Output

A detailed, organized, and easy-to-understand explanation.

### Comparison

| Version   | Quality   |
|-----------|-----------|
| Version 1 | Basic     |
| Version 2 | Better    |
| Version 3 | Good      |
| Version 4 | Excellent |

### 13. Real-World Applications of Prompt Engineering

#### Education

- Study notes
- Quiz generation
- Assignment help

#### Software Development

- Code generation
- Debugging
- Documentation

#### Business

- Email writing
- Report generation
- Meeting summaries

#### Healthcare

- Medical report summarization
- Clinical documentation

#### Marketing

- Product descriptions
- Blog writing
- Social media content

## Human Resources

- Resume review
- Job description generation
- Interview question preparation

## 14. Advantages

- Improves AI response quality
- Saves time
- Increases productivity
- Generates structured outputs
- Automates repetitive tasks

## 15. Limitations

- Depends on prompt quality
- May produce incorrect information
- Requires human review for critical tasks
- Long prompts consume more tokens

## 16. Interview Questions

### Q1. What is Prompt Engineering?

Answer: Prompt Engineering is the process of designing clear and effective prompts to obtain accurate and useful responses from AI models.

### Q2. What is a token?

Answer: A token is the smallest unit of text processed by an AI model.

### Q3. What is Temperature?

Answer: Temperature controls the creativity and randomness of AI-generated responses.

### Q4. What is Zero-shot Prompting?

Answer: Zero-shot prompting asks the AI to perform a task without providing examples.

### Q5. What is Few-shot Prompting?

Answer: Few-shot prompting provides multiple examples before asking the AI to perform a similar task.

### Q6. Why are API Keys important?

Answer: API keys authenticate users, secure API access, and track usage.

### Q7. Name two applications of Prompt Engineering.

Answer:

- Code generation
- Content creation

## 17. Conclusion

Prompt Engineering is an essential skill for effectively using Large Language Models. Well-designed prompts improve the quality, relevance, and accuracy of AI-generated responses. Understanding concepts such as tokens, temperature, prompt types, and API keys enables users to use AI more efficiently.

As AI continues to evolve, Prompt Engineering will remain a valuable skill in education, software development, healthcare, business, and many other industries.

## Final Summary

This document covered:

- Large Language Models (LLMs)
- Prompt Engineering
- Components of a Good Prompt

- Tokens
- Temperature
- Prompt Types
- API Keys
- Best Practices
- Common Mistakes
- Prompt Improvement
- Applications
- Advantages and Limitations
- Interview Questions
- Conclusion

These concepts provide a strong foundation for using AI tools effectively and responsibly.